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EUROPEAN UNION AVIATION SAFETY AGENCY SAFETY INFORMATION BULLETIN
EASA SIB No.: 2024-09 Issued: 14 October 2024
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Subject: Mitigation of Ramp Collision Risks and Ground Vehicle Speed Management on Aprons

Ref. Publications:

- **Regulation (EU) No 139/2014** (Aerodromes Regulation):
  - Annex III (**Part-ADR.OR**): Points ADR.OR.D.017 and ADR.OR.D.027.
  - Annex IV (**Part-ADR.OPS**): Points ADR.OPS.B.025, ADR.OPS.B.030, and ADR.OPS.B.070.
- **Regulation (EU) No 376/2014** on the reporting, analysis, and follow-up of occurrences in civil aviation.
- **ICAO Doc 9137** (*Airport Services Manual*), Part 8 — *Airport Operational Services*.
- **European Plan for Aviation Safety (EPAS)** — Safety Priority *Ground Safety / Apron Operations*.

Applicability:

- National Competent Authorities (NCAs / CAAs).
- Aerodrome Operators holding certificates under Regulation (EU) No 139/2014.
- Ground Handling Service Providers (GHSPs).
- Apron Management Service (AMS) units and Air Traffic Services (ATS) providers.

1. Description & Operational Context

Analysis of safety data reported to the European Central Repository (ECR) under Regulation (EU) No 376/2014 indicates a recurring trend of ground collisions and near-miss occurrences on aerodrome aprons. A significant proportion of these occurrences involve **Ground Support Equipment (GSE)** and service vehicles colliding with parked or taxiing aircraft, passenger boarding infrastructure, and other vehicles.

Contributing factor analyses reveal that **excessive driving speed**, failure to yield right-of-way, reduced visual scan patterns during high-workload turnaround phases, and degraded situational awareness in adverse weather or night operations represent primary precursors to apron incursions and structural aircraft impact.

2. Safety Concern

High-energy vehicle impacts in the ramp environment pose immediate safety risks:

- **Unreported Aircraft Structural Damage:** High-speed GSE impacts with aircraft fuselages, radomes,engine cowlings, or control surfaces have on several occasions gone undetected or unreported by ground staff, leading to aircraft departing with compromised structural integrity.

- **Apron Incursions into Taxiways:** Vehicles operating above designated speed thresholds have failed to stop prior to entering active aircraft taxi lanes, compromising separation with taxiing aircraft.

- **Personnel Hazard:** Increased vehicle speeds inside congested Equipment Clearance Areas (ECA) and Equipment Staging Areas (ESA) correlate with severe ground personnel injuries during aircraft pushback and loading operations.

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3. Regulatory Baseline (Part-ADR.OPS.B.025)

Pursuant to point ADR.OPS.B.025 (*Operation of vehicles*), the aerodrome operator must establish and verify rules governing the movement and speed of all vehicles operating on the movement area. This includes an **Apron Driving Authorisation (ADA)** scheme, mandatory driver training syllabi, vehicle roadworthiness checks, and formal enforcement mechanisms.

4. Recommendations

EASA recommends that **Aerodrome Operators**, in coordination with **Ground Handling Service Providers (GHSPs)** and **Competent Authorities (e.g., DSAC)**, take the following actions:

A. Infrastructure Design, Speed Limits & Zoning

1. **Differentiated Speed Limits:** Establish clear, enforceable speed regimes tailored to apron zones:
    - **General Apron Roadways:** Typically not exceeding **25–30 km/h**.
    - **Aircraft Stand / Servicing Zones:** Strictly limited to **5–10 km/h** (walking pace) in the vicinity of parked aircraft.
    - **Baggage Make-up & Tunnel Facilities:** Capped at **10–15 km/h** with mandatory blind-corner mirrors.
2. **Visual Markings & Road Signage:** Audit movement area road markings, ensuring speed limit roundels,stop bars, yield markings, and pedestrian crossings conform strictly to CS-ADR-DSN visual standards.

3. **Telematics & Speed Limiting:** Promote the adoption of physical/electronic speed governors on GSE and support vehicles, as well as GPS geofencing that automatically restricts top speed when entering designated aircraft stands.

B. Training, Authorisation & "Just Culture" Enforcement

1. **Apron Driving Authorisation (ADA) Rigour:** Ensure driver licensing syllabi (ADR.OR.D.017) include mandatory practical assessments on stopping distances of loaded tractor-trailer trains and blind-spot geometry around widebody aircraft.

2. **Radar & Telematics Surveillance:** Deploy speed-detection devices (e.g., mobile radar units, fixed speed cameras, or telematics tracking) on major apron roads to identify non-compliant operators.

3. **Transparent Reporting Culture:** Reiterate that low-speed vehicle contacts with aircraft (even superficial scrapes) must be reported immediately without punitive fear, ensuring maintenance inspections are completed prior to flight release.

C. Stakeholder Coordination via the Ramp Safety Committee (RSC)

1. Convene regular meetings of the local **Apron / Ramp Safety Committee** to review radar surveillance data, vehicle telematics trends, and ramp collision near-miss reports.

2. Establish joint safety audits between the aerodrome operator and tenant ground handlers to ensure third-party contractors adhere to the vehicle speed rules published in the Aerodrome Manual.

5. Status of this SIB

- This SIB is **informative and non-mandatory**. It does not constitute an EU regulation or an EASA Certification Specification.

- At this time, the safety condition described does not warrant a Safety Directive (SD) under Regulation (EU) No 139/2014, Annex II, ADR.AR.A.040.

Contact:

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